화학특강 (2010년 2학기)

제4장 고분자, Dendrimer 그리고 Supramolecular Chemistry

황현석

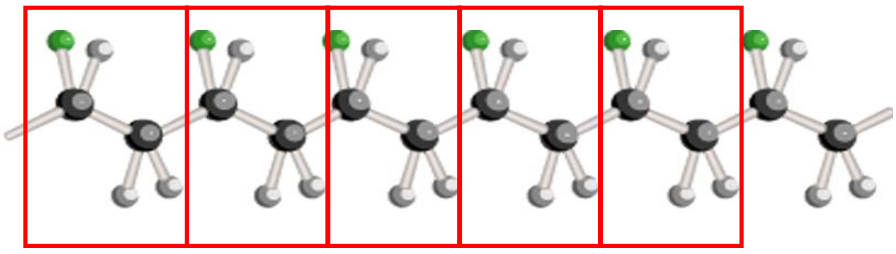
강원대학교 화학과

2010년 10월 18일 월요일

Polymers

Greek: “polys” ~ many
“meros”~parts

Polymers are molecules consisting of a (large) number of structural units called monomers, that are connected by covalent bonds.



The monomers are small (organic) molecules, linked together in a process called polymerization.



Chang, 25.1

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Molar Mass Distributions of Polymers

- Degree of polymerization (N): the number of monomers in a polymer
- Monodisperse and polydisperse
  - If all polymers in a sample have the same number of monomers, the sample is **monodisperse**.
  - If polymers in a sample consist of a various number of monomers, the sample is **polydisperse**.
- Number-average molar mass:

$$M_n = \sum_N n_N M_N$$

- Weight-average molar mass:

$$M_w = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} = \frac{\sum_N n_N M_N^2}{M_n}$$

- Polydispersity index

$$I_p = \frac{M_w}{M_n}$$



(예제) 몰질량이 1.0×10^5 g/mol인 gelatin monomer 단백질과 몰질량이 2.0×10^5 g/mol인 gelatin dimer 단백질을 포함하는 시료가 있다. gelatin monomer의 몰분율이 0.5라고 할 때, 시료 속 gelatin 단백질의 number-average 몰질량과 weight-average 몰질량을 구하여라. 이 단백질의 polydispersity index는 얼마인가?

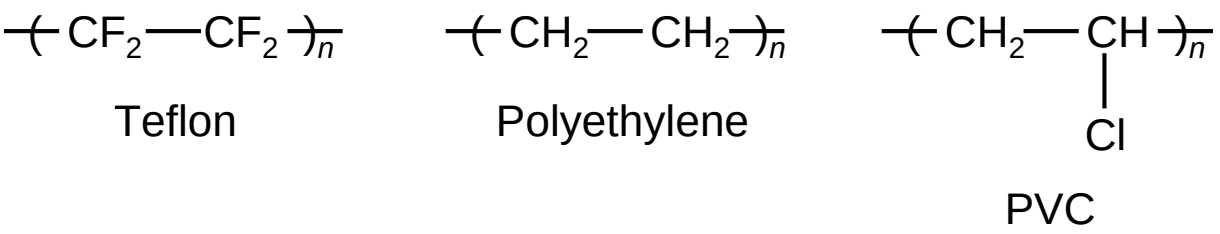
$$\text{number-average molar mass} = \sum_N n_N M_N = 1.5\times10^5 \text{ g/mol}$$

$$\text{weight-average molar mass} = \frac{\sum_N n_N M_N^2}{\sum_N n_N M_N} \cong 1.67\times10^5 \text{ g/mol}$$

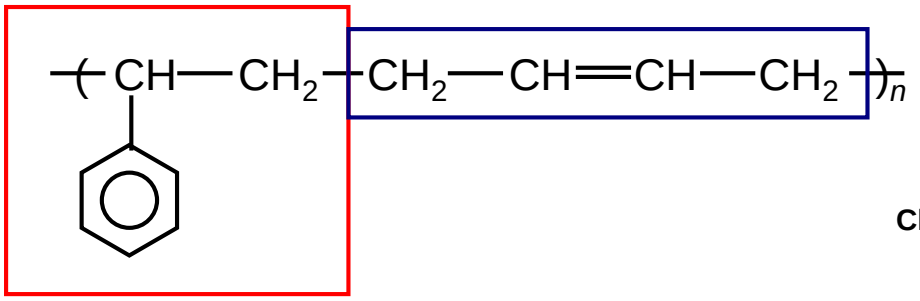
polydispersity index $I_p = \frac{M_w}{M_n} = \frac{10}{9}$

Homopolymers and Copolymers

- ***Homopolymer*** is a polymer made up of only one type of monomer



- ***Copolymer*** is a polymer made up of two or more monomers.



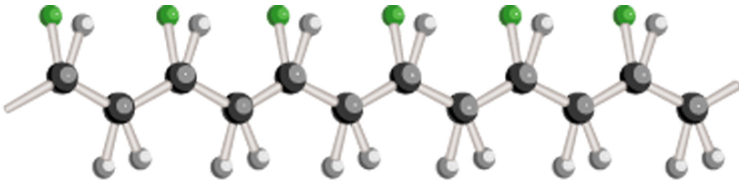
Chang, 25.1

Styrene-butadiene rubber

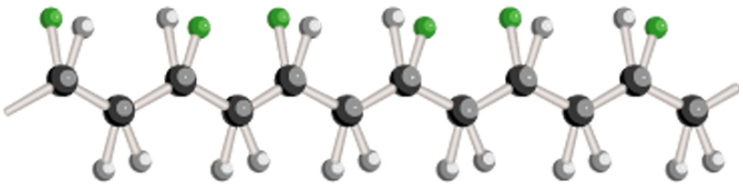
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Stereoisomers of Polymers

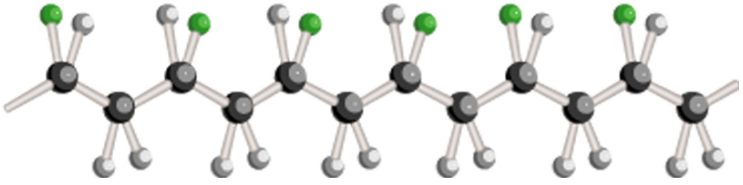
Isotactic: similar side groups always on same side of the chain. Highest melting point and great crystallinity



Syndiotactic: side groups on alternating sides of the chain



Atactic: side groups disposed at random. Rubbery,amorphous, and relatively weak



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Common Polymers (1)

TABLE 25.1 Some Monomers and Their Common Synthetic Polymers

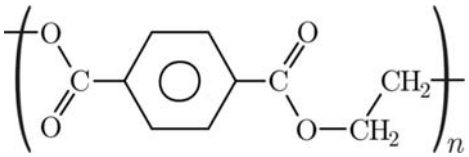
Monomer		Polymer	
Formula	**Name**	**Name and Formula**	**Uses**
$H_2C{=}CH_2$	Ethylene	Polyethylene $-(CH_2-CH_2)-_n$	Plastic piping, bottles, electrical insulation, toys
$H_2C{=}C(H)(CH_3)$	Propylene	Polypropylene $-(CH(CH_3)-CH_2-CH(CH_3)-CH_2)-_n$	Packaging film, carpets, crates for soft-drink bottles, lab wares, toys
$H_2C{=}C(H)(Cl)$	Vinyl chloride	Poly(vinyl chloride) (PVC) $-(CH_2-CH(Cl))-_n$	Piping, siding, gutters, floor tile, clothing, toys
$H_2C{=}C(H)(CN)$	Acrylonitrile	Polyacrylonitrile (PAN) $-(CH_2-CH(CN))-_n$	Carpets, knitwear

Chang, 25.2

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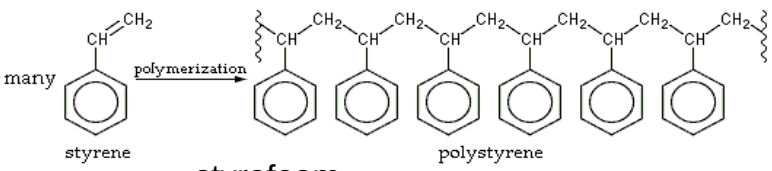
Common Polymers (2)

- Polyethylene terephthalate (PET)





- Polystyrene





styrofoam



- High-density polyethylene (HDPE) and low-density polyethylene (LDPE) and





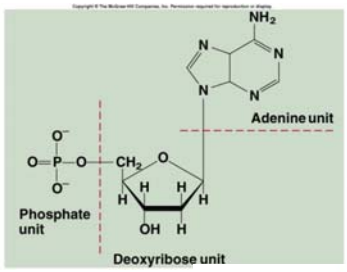




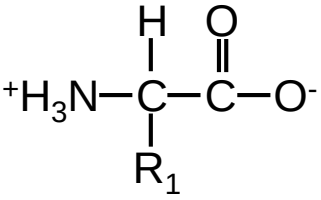
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Biopolymers

- **DNA (RNA)** consists of subunits called nucleotides: sugar deoxyribose (ribose), phosphate group PO_4^- and a base.

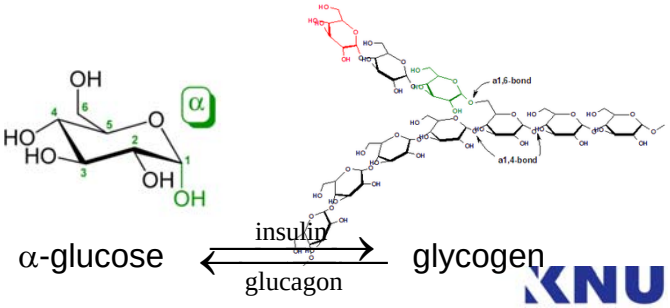


- **Proteins** are polypeptides with 20 different amino acids as the structural subunits.





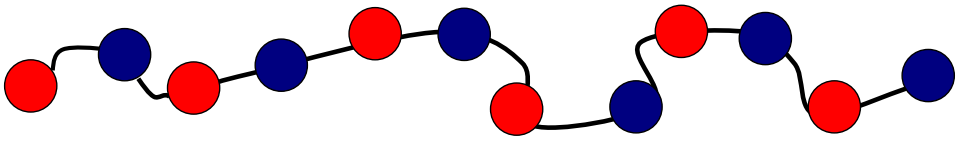
- **Polysaccharides** are complex, often branched, carbohydrates, consisting of monosaccharides (simple sugars)



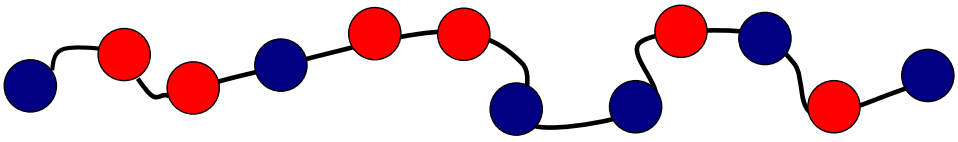
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Copolymers

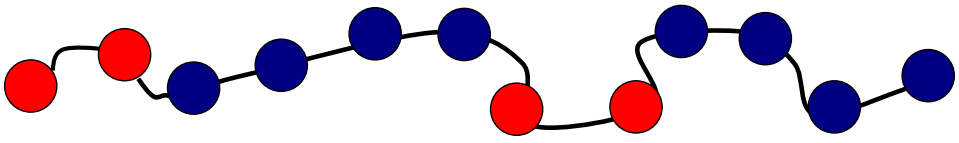
- Alternating copolymers



- Random copolymers



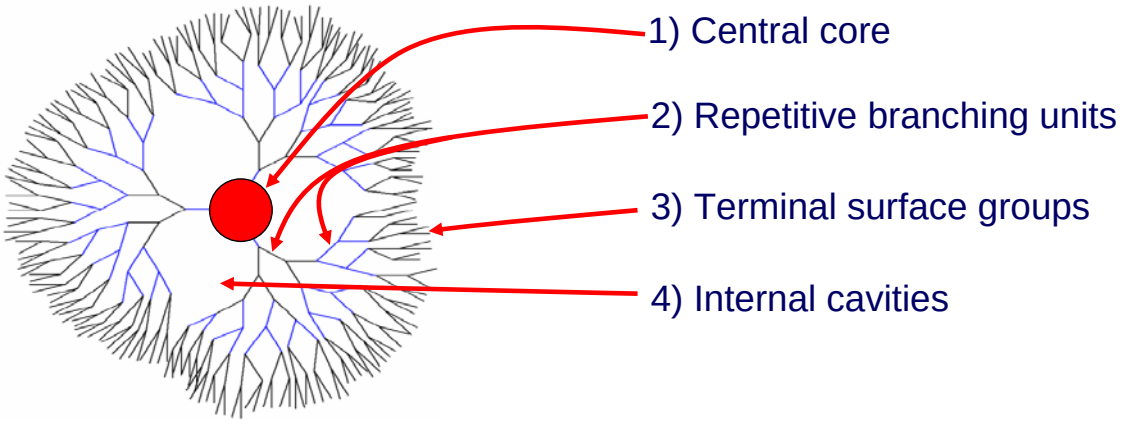
- Block copolymers



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Dendrimers (1)

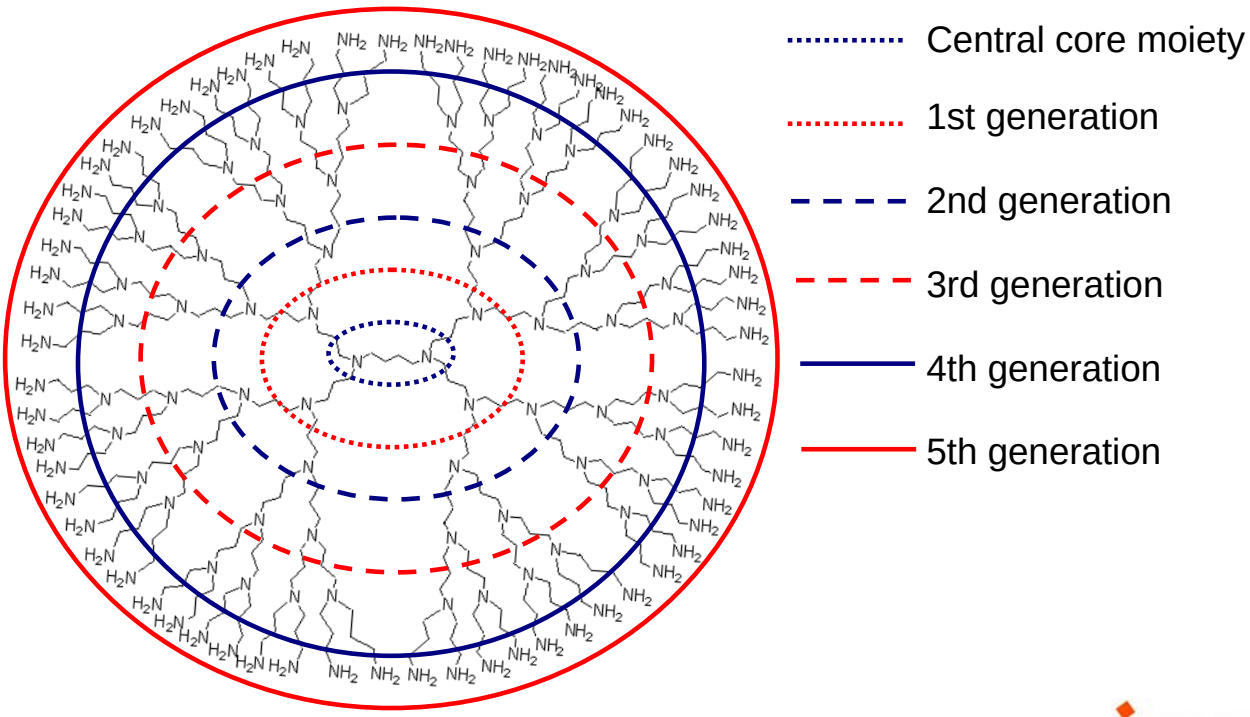
- Dendrimers: highly branched (hyperbranched) polymers that have a well defined structure. [“dendros” (Greek) ~ tree, branch]



- Divergent synthesis: from the core to the surface
- Convergent synthesis: from the surface to the core

Dendrimers (2)

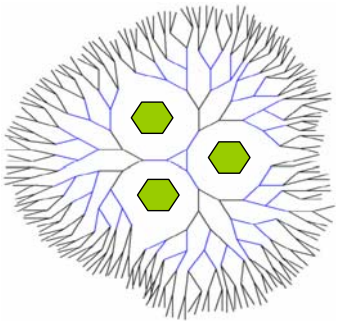
- Polyamidoamin (PAMAM) dendrimers



Dendrimers (3)

- Properties of dendrimers are dominated by functional groups on the surface of the globular molecule. For example, a dendrimer can be made water-soluble with hydrophilic surface groups
  (e.g. R-COOH $\rightarrow$ R-COO$^-$ + H$^+$)

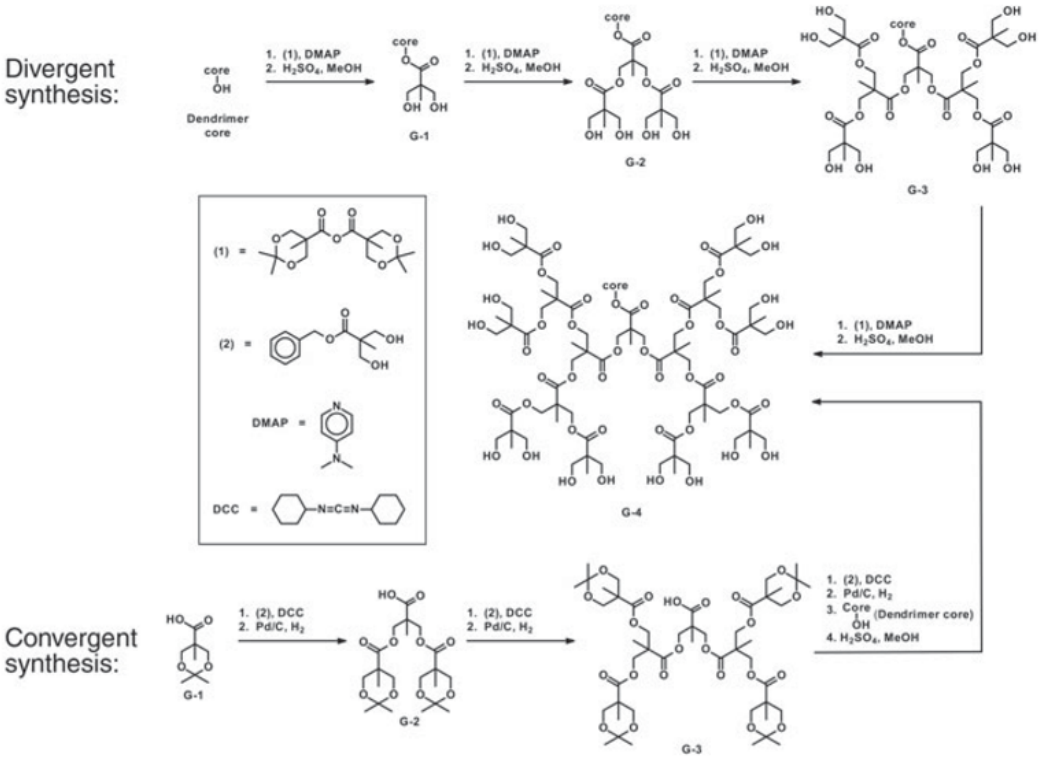
- Dendrimers with hydrophobic internal structure can act as drug carriers. Alternatively, the drug could be attached directly to the dendrimer surface groups.



= encapsulated drug

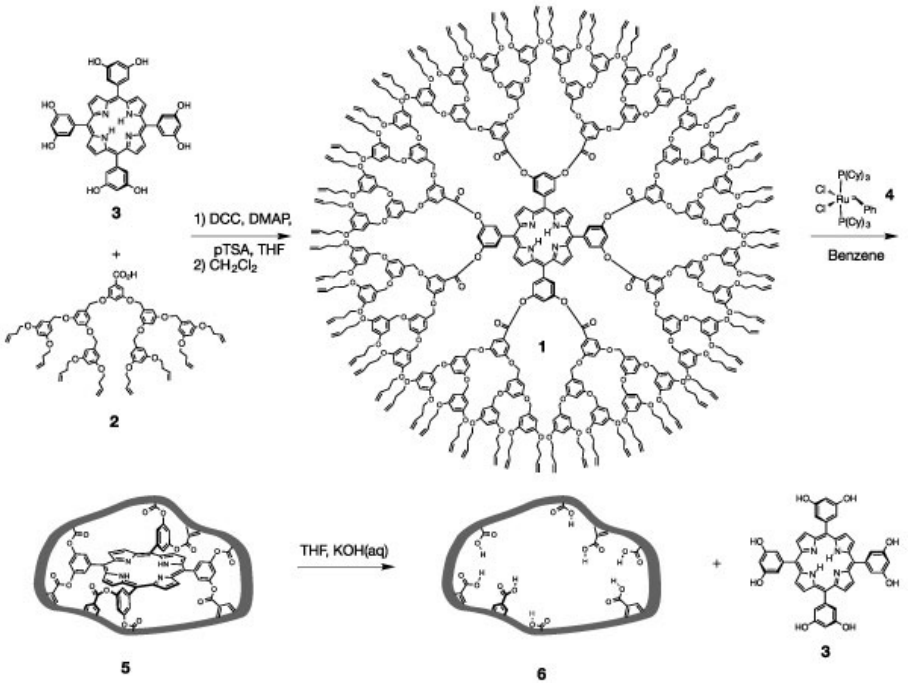
- Dendrimers have also shown promise in the development of organic LEDs: photons can be directly toward a luminescent core moiety by suitably chosen surface groups.

Dendrimer Synthesis via Divergent and Convergent Approaches



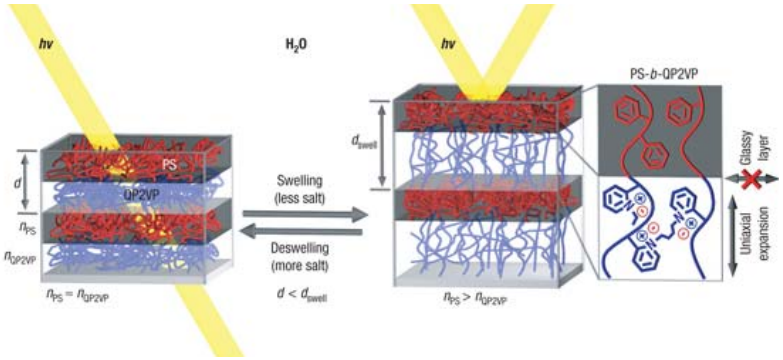
Lee et al., Nature Biotechnology **23**, 1517 (2005)

Synthetic Hosts by Molecular Imprinting

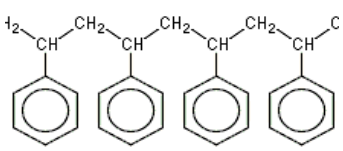


Zimmerman et al., Nature **34**, 399 (2002)

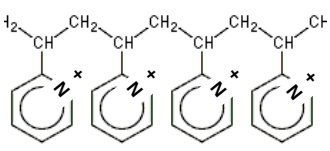
Block-Copolymer Photonic Gel







polystyrene
(=PS)

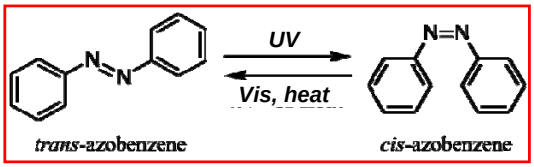


quarternized poly(2-vinyl pyridine)
(=QP2VP)

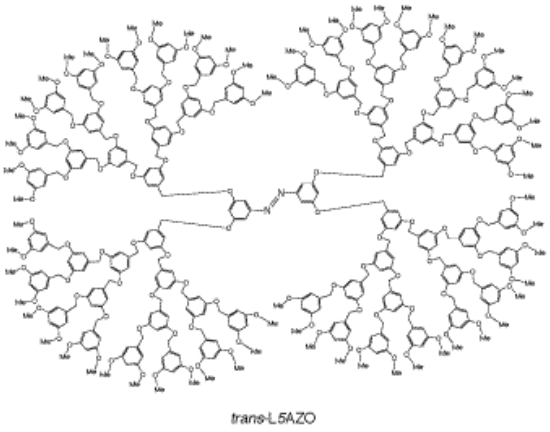
Kang et al., Nature Materials **6**, 957 (2007)

Photoisomerization in dendrimers by harvesting of low-energy photons

- Cis-trans isomerization of azobenzene



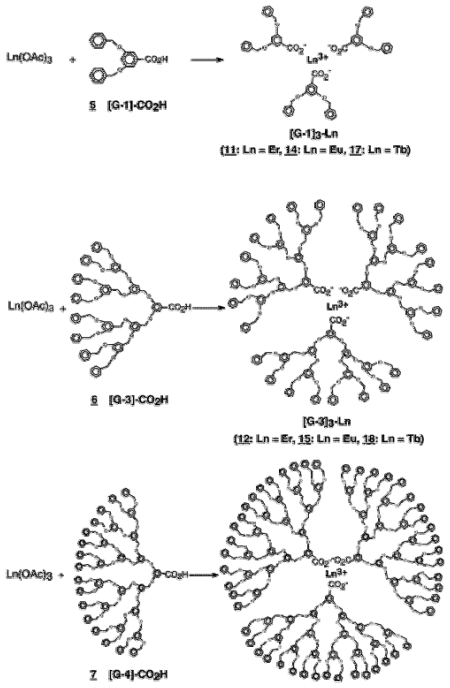
- Photoactive dendrimer system



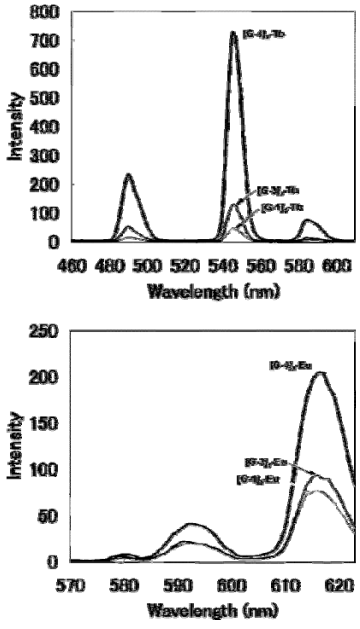
Fischer et al., Angew. Chem. Int. Ed. **38**, 884 (1999)

Luminescence Properties of Lanthanide Ions through Site-Isolation and Antenna Effects

- Ln-cored dendrimer complexes



- Emission spectra of Ln-cored dendrimer complexes



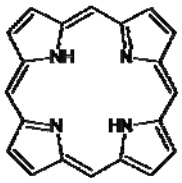
Kawa and Fréchet, Chemistry of Materials **10**, 286 (1998)

Supramolecular Chemistry

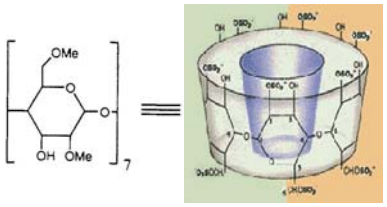
- **Supramolecular chemistry:** the area of chemistry that focuses on the noncovalent intermolecular interactions of molecules.
- Kinds of noncovalent intermolecular interactions
  - Hydrogen bonding
  - π-π stacking interaction
  - Hydrophobic interaction
  - Electrostatic interaction
    a. Charge-dipole interaction
    b. Dipole-dipole interaction
  - van der Waals interaction

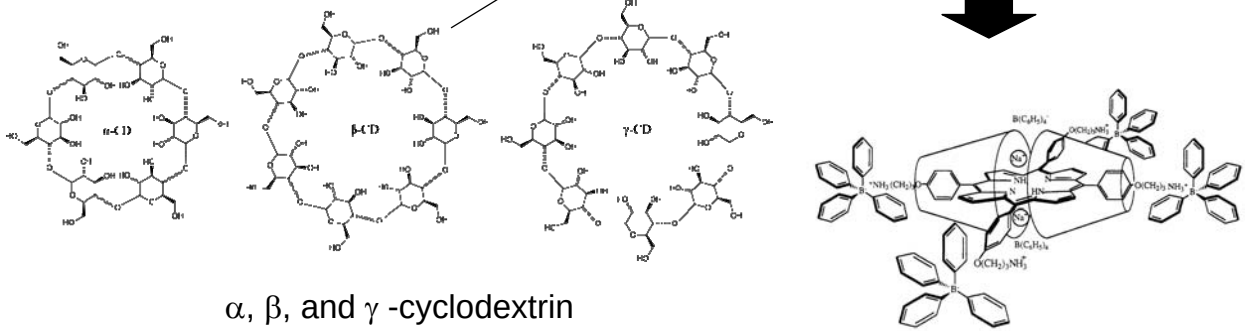
Supramolecular Chemistry: Hydrophobic Interaction

- Supramlecular complex containing porphyrin and β-cyclodextrin (β-CD) → hydrophobic interaction



porphyrin

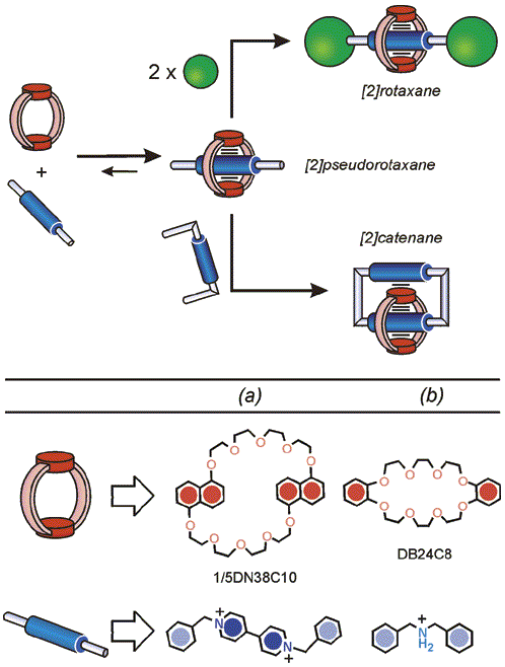




α, β, and γ -cyclodextrin

Manka and Lawrence, JACS **112**, 2440 (1990)

Interlocked Molecules



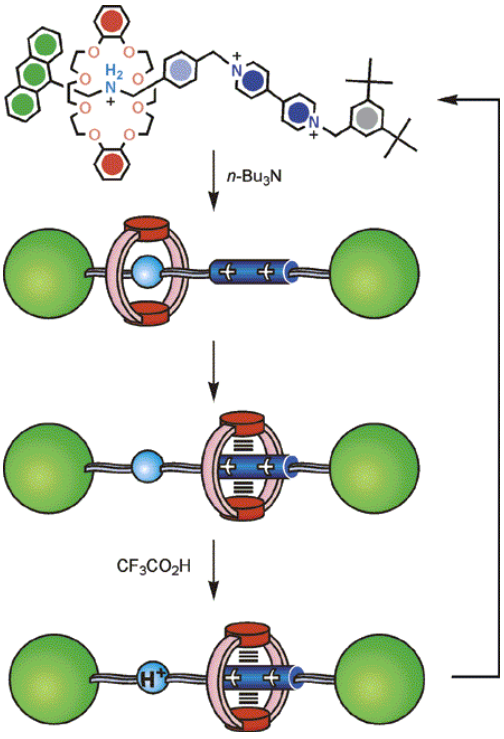
- **Rotaxane**: molecular assemblies in which a linear component is mechanically threaded through one or more cyclic molecules with the ends of the linear component terminated with bulky groups.
- **Catenane**: linked supramolecular systems consisting of cyclic subunits held by a mechanical means
- **Host guest adduct**: Interaction between π-electron rich hydroquinone and π-electron deficient pyridinium to form π stack (3.7 Å between rings). dispersive, electrostatic interactions.

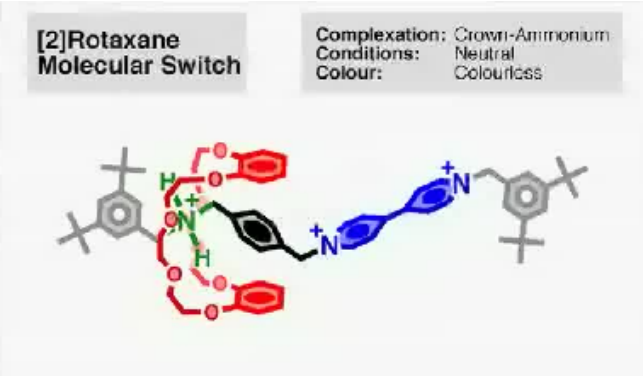
Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

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Rotaxanes: A Molecular Shuttle

"rot" means "wheel and axa means " axis

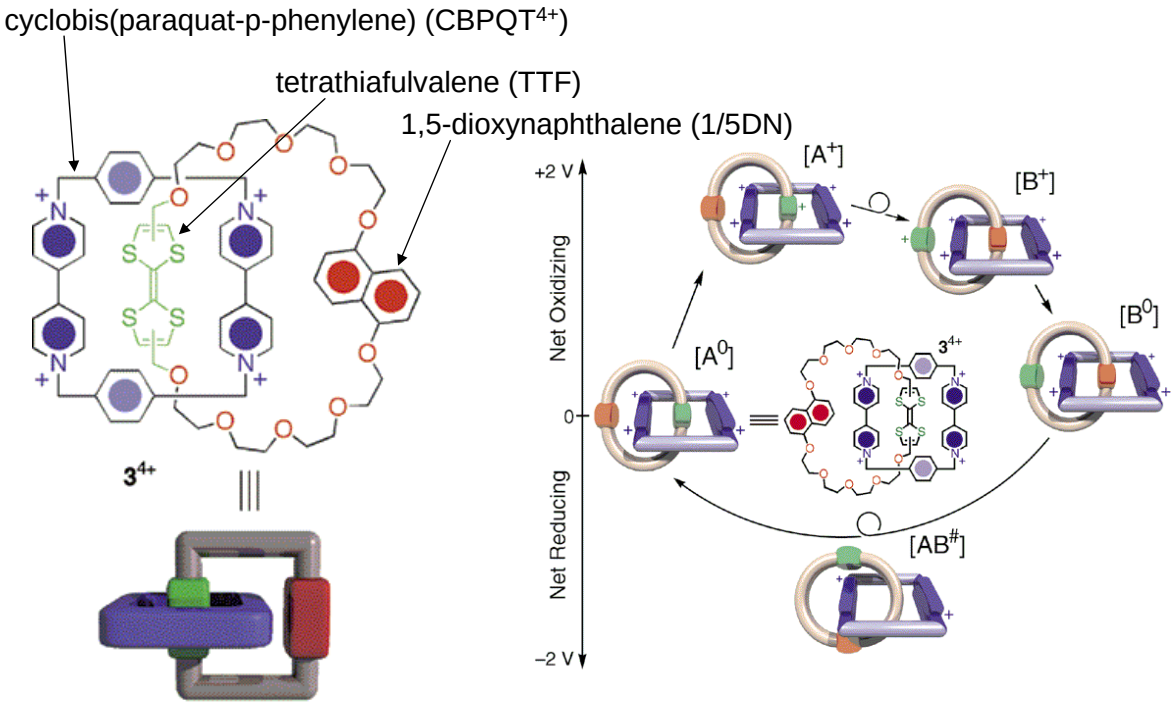




Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

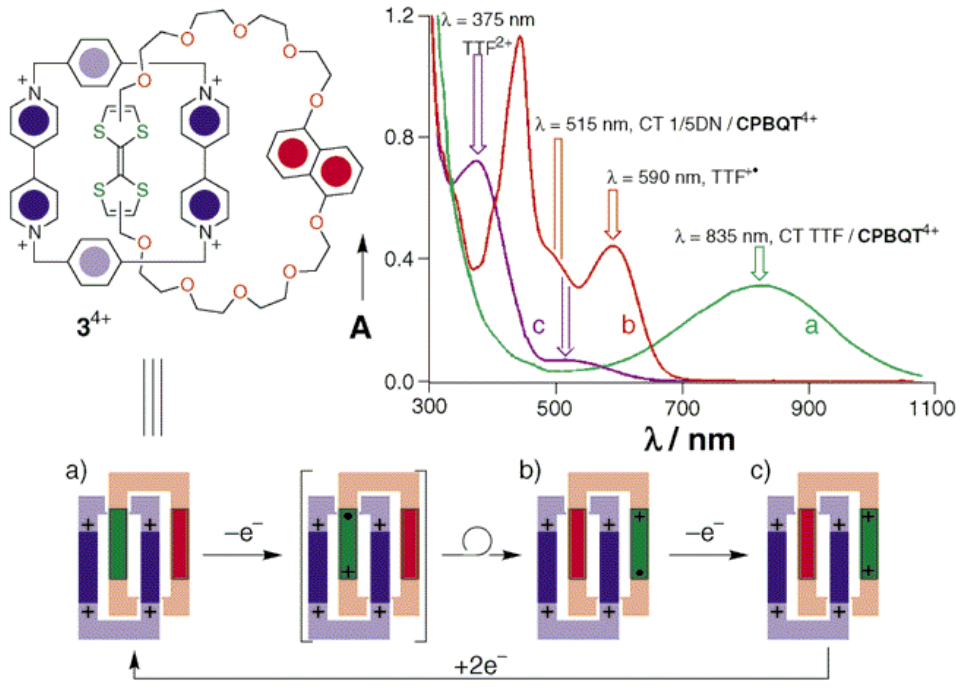
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Catenanes: A Molecular Switch



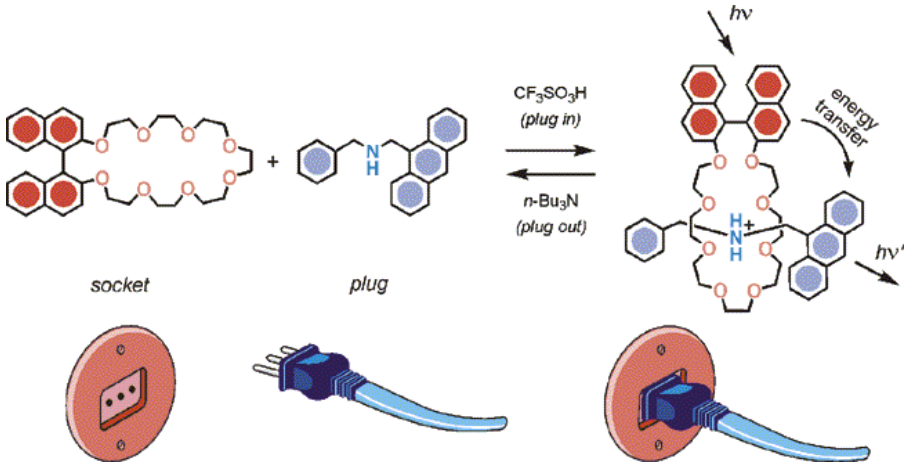
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

Catenanes: A Molecular Switch



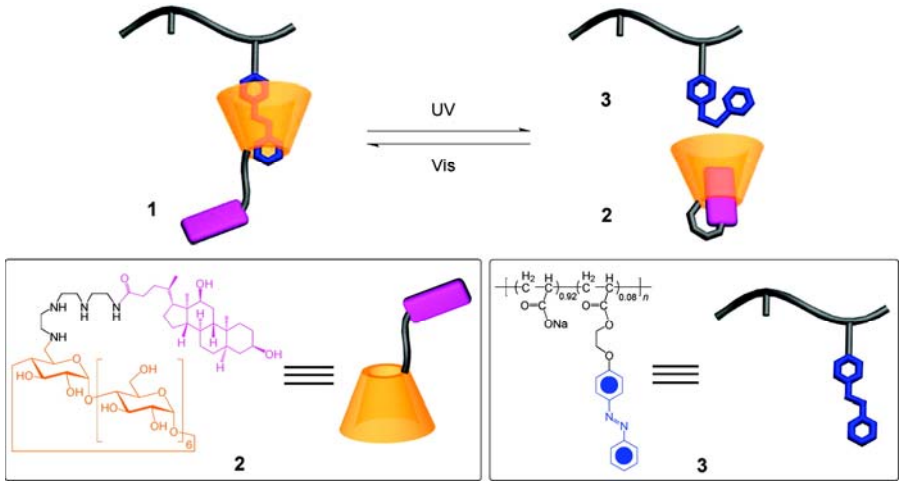
Balzani et al., Acc. Chem. Res. **34**, 445 (2001); Stoddart et al., Acc. Chem. Res. **34**, 433 (2001)

A Molecular Socket-Plug device



Balzani et al., Acc. Chem. Res. **34**, 445 (2001)

Azobenzene-Based Light-Responsive Hydrogel System



Colloids (1)

- Colloids: a dispersion of particles of one substance (the dispersed phase) throughout a dispersing medium maid of another substance.

TABLE 12.4 Types of Colloids

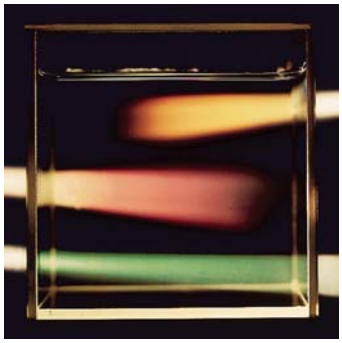
Dispersing Medium	Dispersed Phase	Name	Example
Gas	Liquid	Aerosol	Fog, mist
Gas	Solid	Aerosol	Smoke
Liquid	Gas	Foam	Whipped cream
Liquid	Liquid	Emulsion	Mayonnaise
Liquid	Solid	Sol	Milk of magnesia
Solid	Gas	Foam	Plastic foams
Solid	Liquid	Gel	Jelly, butter
Solid	Solid	Solid sol	Certain alloys (steel), gemstones (glass with dispersed metal)

Chang, General Chemistry **12.8**

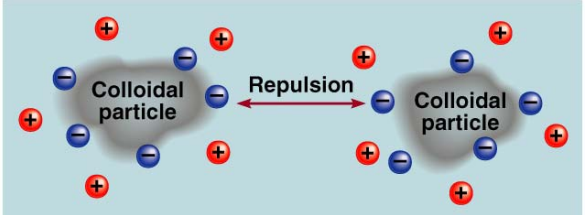
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Colloids (2)

- Tyndall effect



- Electrostatic repulsion

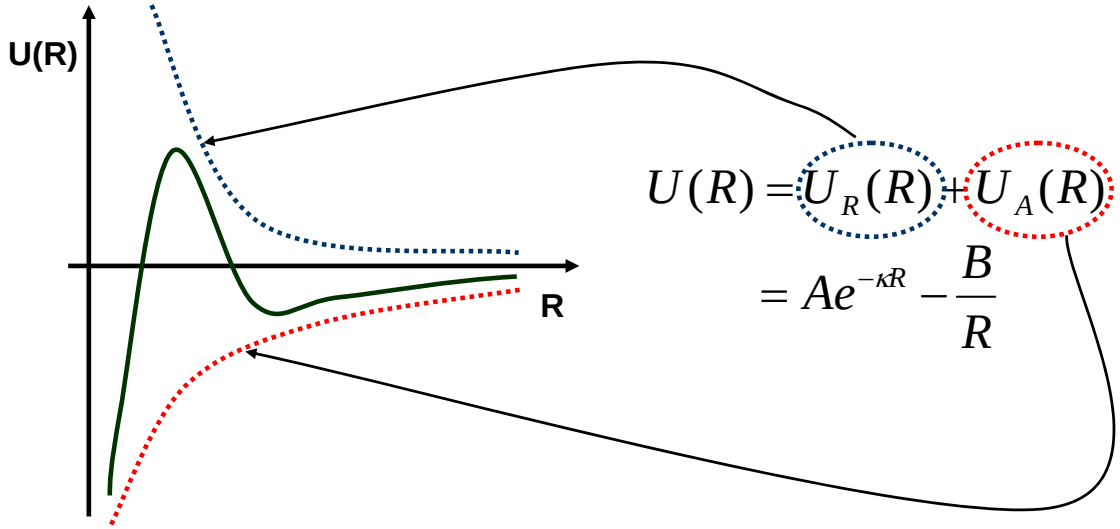


Chang, General Chemistry **12.8**

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Colloids (3)

- Interactions between colloids



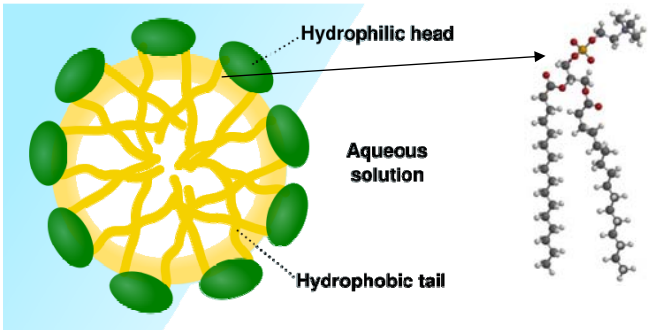
- Coagulation: the destruction of a colloid

Chang, General Chemistry **12.8**

Colloids (4)

- Mayonnaise





- River Delta

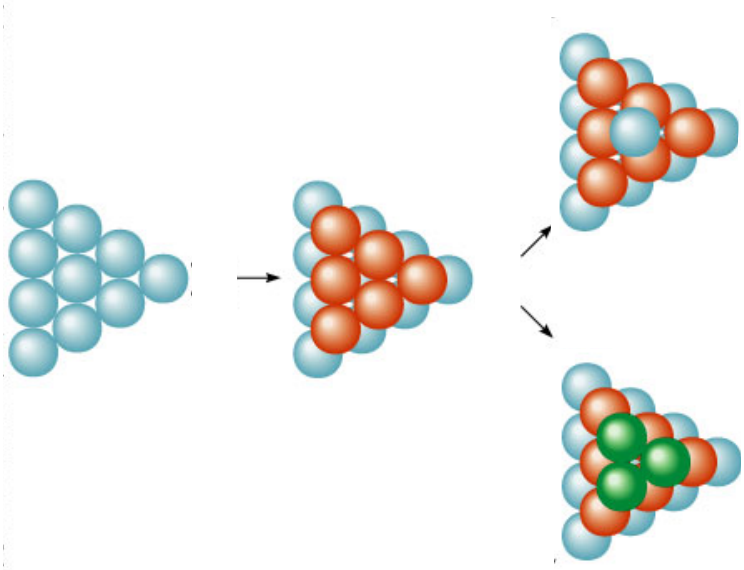


Nile Delta

Chang, General Chemistry **12.8**

Colloids and Crystallization

- Hexagonal and cubic closed packed structures



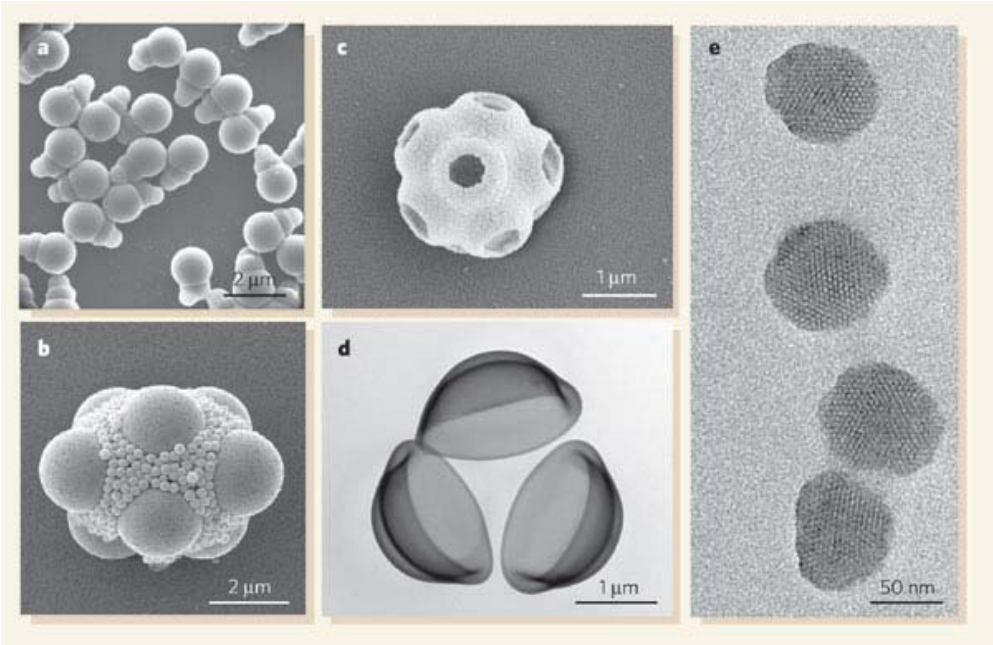
hexagonal closed packed structure (aba)

cubic closed packed structure (abc)

Chang, General Chemistry **11**.

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Colloids in the Nanotechnology

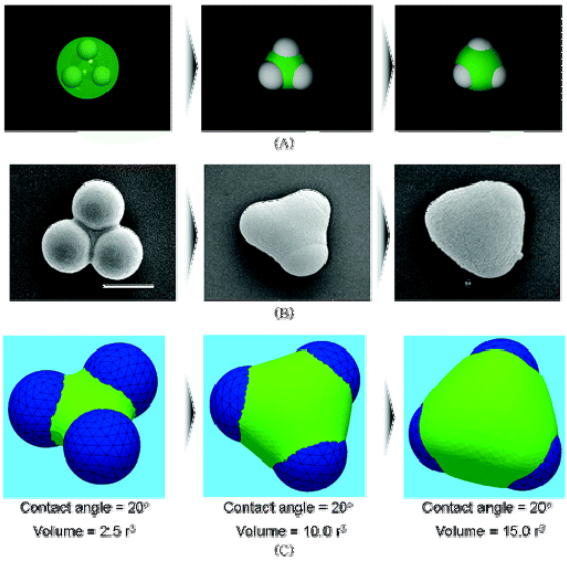


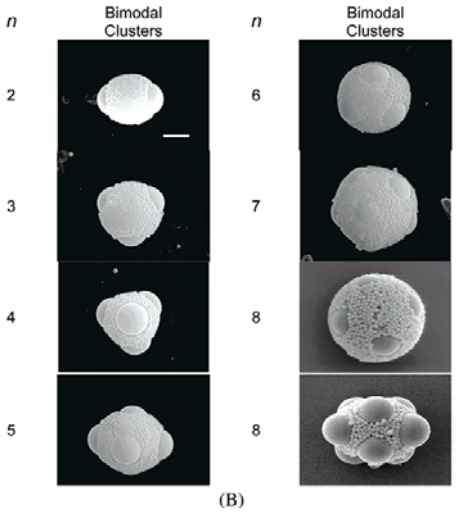
van Blaaderen, Nature **439**, 545 (2006)

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Colloids in the Nanotechnology

- Formation of a trimer cluster of silica microparticles with silica nanoparticles from water-in-oil emulsions.

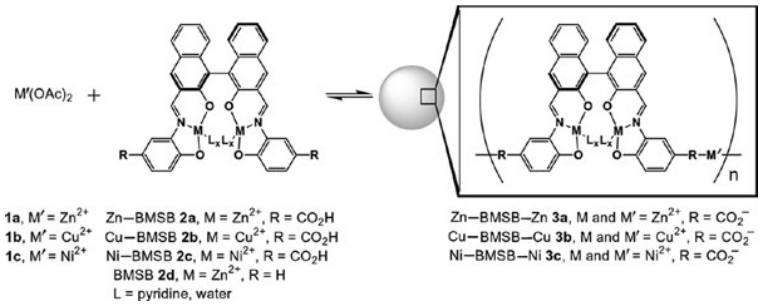


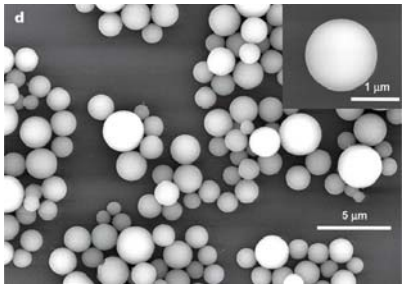


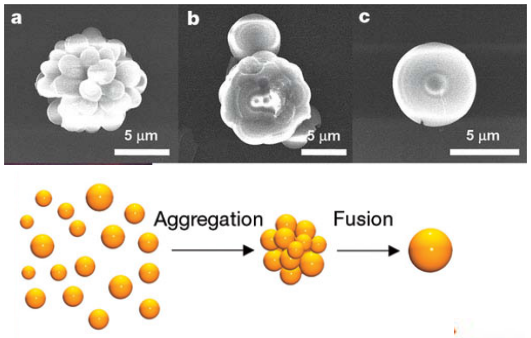
Young-Sang Cho, J. Am. Chem. Soc. **127**, 15968 (2005)

Colloids in the Nanotechnology

- assembly of metal ions and homochiral carboxylate-functionalized binapthyl bis-metallo-tridentate Schiff base (BMSB) building blocks



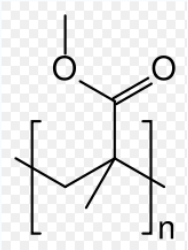




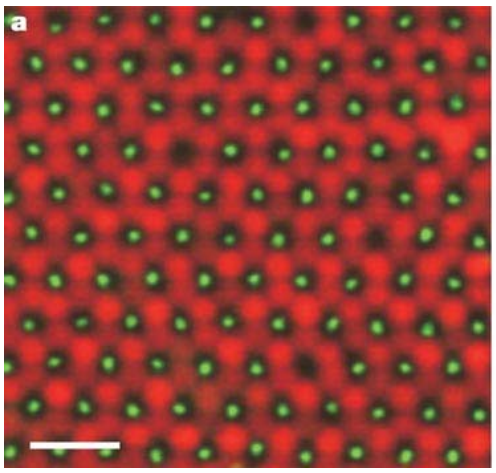
Oh and Mirkin, Nature **431**, 651 (2005)

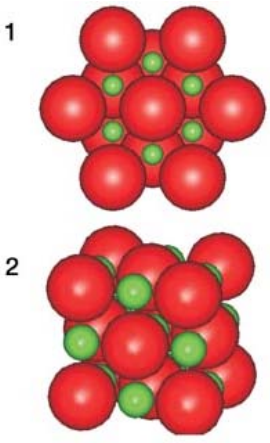
Colloids and Crystallization

- Poly(methyl methacrylate) (PMMA) colliods



- Crystallization of PMMA colloids





Leunissen, Nature **437**, 235 (2005)